

Emotion Detection Using Voice

Project Report

Machine Learning (UML501)

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Introduction

Emotion is a fundamental component of human communication. Apart from linguistic content, speech carries a rich set of paralinguistic cues that indicate the emotional state of the speaker. Detecting emotions from speech has become an essential research area in machine learning, signal processing and human–computer interaction.

With the rise of intelligent voice assistants, mental health monitoring systems, smart classrooms, and customer-care automation, understanding emotions from voice signals has become increasingly important. Unlike text, speech contains pitch, intensity, rhythm, spectral features, and temporal variations that make it a highly informative and challenging medium.

This project proposes a deep learning–based emotion recognition system using **MFCC (Mel-Frequency Cepstral Coefficients)** for feature extraction and **Long Short-Term Memory (LSTM)** networks for sequential learning. LSTMs are powerful models capable of capturing temporal dependencies in audio signals, making them ideal for speech emotion detection.

Our dataset, **HSED (Human Speech Emotion Dataset)**, was custom recorded by students in our college, ensuring real-world relevance.

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Need of the Problem

As automation becomes widespread, emotion-aware systems are required for natural, empathetic and context-sensitive interaction. Some key motivations include:

- **Human–Computer Interaction:** Systems like Alexa, Google Assistant, and Siri need emotional context to give appropriate responses.
- **Mental Health Monitoring:** Emotional irregularities in speech can indicate stress, loneliness, or early psychological disturbances.
- **Customer Care Analytics:** Call centers rely on emotion analysis to improve customer satisfaction.
- **Education:** Intelligent tutoring systems can adjust difficulty levels based on a student’s emotional state.
- **Security:** Emotion detection can assist in lie detection and behavioral analysis.

Traditional rule-based or statistical approaches struggle with the complexity of speech signals. Deep learning models, especially LSTMs, show significant improvement in accuracy and robustness.

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Problem Statement

To design, implement and evaluate a deep learning model capable of classifying emotional states from audio speech signals by using MFCC-based feature extraction and an LSTM neural network architecture.

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Objectives and Existing Approaches

Objectives

- To collect a speech emotion dataset in a real, unconstrained environment.
- To preprocess raw audio and extract MFCC features.
- To design and train an LSTM-based neural model.
- To compare training and testing performance.
- To predict the emotion of new unseen audio samples.

Existing Approaches

1. Classical Machine Learning:

- SVM (Support Vector Machine)
- KNN (K-Nearest Neighbors)
- Random Forest

Limitations include inability to capture temporal dependencies in audio.

2. Deep Learning-Based Approaches:

- Convolutional Neural Networks (CNN)
- Recurrent Neural Networks (RNN)
- Hybrid CNN-LSTM models

LSTMs outperform classical ML models because they preserve memory across time steps, making them well-suited for speech patterns.

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Dataset Description

We collected our own dataset named **HSED (Human Speech Emotion Dataset)**. Students recorded speech samples using mobile phones in varied indoor environments.

Dataset Characteristics

- Total samples: multiple .wav files across emotion categories.
- Categories (example):
 - Happy
 - Sad
 - Angry
 - Neutral
- Sampling rate: varies (handled during preprocessing).
- File format: .wav.
- Real, noisy environment reflecting practical conditions.

Unlike curated datasets such as RAVDESS, our dataset provides practical challenges due to background noise and varying

recording distances.

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Methodology / Model Description

The project workflow consists of the following stages:

1. Data Extraction

The dataset ZIP file was validated, extracted, and recursively scanned to collect all audio files.

2. MFCC Feature Extraction

We extracted 40 MFCC coefficients from each audio file using the `librosa` library. MFCCs closely represent human auditory perception and serve as the most widely used features in speech processing.

3. Sequence Padding

Since audio files vary in duration, we padded MFCC sequences using zero-padding to a fixed maximum length.

4. Label Encoding

Emotion labels were converted to numeric form using LabelEncoder.

5. Train-Test Split

Data was split into:

- 80% training
- 20% testing

6. LSTM Model Architecture

- LSTM (128 units, return_sequences=True)
- Dropout (0.3)
- LSTM (128 units)
- Dropout (0.3)
- Dense (output classes, softmax activation)

7. Training

The model was trained for 50 epochs with batch size 32.

8. Prediction

A random audio sample was selected from the dataset and passed to the model for real-time emotion prediction.

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Implementation Details

Our code performs the following major steps:

- Imports required libraries: librosa, numpy, tensorflow, sklearn.
- Extracts the ZIP dataset and loads all audio files.
- Applies MFCC extraction with error handling.
- Performs padding/truncation to ensure uniform sequence length.
- Converts lists to NumPy arrays.
- Encodes emotion labels using LabelEncoder.
- Builds an LSTM model using tensorflow.keras.
- Compiles the model using the Adam optimizer.
- Evaluates the model on the test data.
- Predicts the emotion of a sample audio file.

The code is modular, readable and suitable for further extension.

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Results and Evaluation

The LSTM model demonstrated strong learning capability and produced the following final performance metrics on the test set:

- **Test Loss:** 0.8798
- **Test Accuracy:** 0.6695 (66.95%)

A test accuracy of nearly 67% is promising given the challenging nature of the dataset, which includes:

- Real-world background noise,
- A relatively small number of samples,
- Natural differences in speaking styles,
- Non-studio recording conditions.

The training process showed smooth convergence, and predictions on unseen audio samples were reasonably accurate. The LSTM layers were able to capture temporal emotional patterns present in the MFCC sequences.

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Conclusion

This project successfully implemented a Speech Emotion Recognition system using MFCC features and LSTM neural networks. Despite the noisy and variable nature of our custom HSED dataset, the model achieved an accuracy of 66.95%, demonstrating the effectiveness of LSTMs in modeling emotional information in speech.

The project met its objectives: data collection, preprocessing, model design, training, evaluation, and prediction. The system forms a solid baseline that can be extended and improved for real-world applications.

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Future Work and Limitations

Limitations

- Limited dataset size and speaker diversity.
- Presence of background noise in many recordings.
- Only a few emotional categories considered.

Future Enhancements

- Collect a larger and more balanced dataset with more speakers.
- Apply noise-reduction and audio-cleaning techniques before MFCC extraction.
- Experiment with hybrid CNN-LSTM or attention-based architectures.
- Deploy the model as a real-time web or mobile application.

References

- Data set used from ELC LEARNING ACTIVITY-EMOTION AWARENESS
- About the model:
https://www.riverpublishers.com/pdf/ebook/chapter/RP_9788770040723C133.pdf